Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.49)

Metric

$$ds^2 = -e^{(-2\,g)} \left(\frac{\partial\,\lambda}{\partial t}\right)^2 \mathrm{d}t \otimes \mathrm{d}t + e^{(2\,\lambda(r,t))} \mathrm{d}r \otimes \mathrm{d}r + r^2 e^{(2\,\lambda(r,t))} \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 e^{(2\,\lambda(r,t))} \sin\left(\theta\right)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$g(t) = g$$

$$\lambda(t, r) = \lambda\left(r, t\right)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} e^{(-g)} \frac{\partial}{\partial t} \lambda\left(r, t\right)\right) dt + \left(-\sqrt{\frac{1}{2}} e^{(\lambda(r,t))}\right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} e^{(-g)} \frac{\partial}{\partial t} \lambda\left(r, t\right)\right) dt + \left(\sqrt{\frac{1}{2}} e^{(\lambda(r,t))}\right) dr$$

$$m = \left(\sqrt{\frac{1}{2}} r e^{(\lambda(r,t))}\right) d\theta + \left(i\,\sqrt{\frac{1}{2}} r e^{(\lambda(r,t))} \sin\left(\theta\right)\right) d\phi$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} r e^{(\lambda(r,t))}\right) d\theta + \left(-i \sqrt{\frac{1}{2}} r e^{(\lambda(r,t))} \sin(\theta)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2} e^{g}}{2 \frac{\partial}{\partial t} \lambda(r,t)}\right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))}\right) dr$$

$$n = \left(-\frac{\sqrt{2} e^{g}}{2 \frac{\partial}{\partial t} \lambda(r,t)}\right) dt + \left(\frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))}\right) dr$$

$$m = \left(\frac{\sqrt{2} e^{(-\lambda(r,t))}}{2 r}\right) d\theta + \left(\frac{i \sqrt{2} e^{(-\lambda(r,t))}}{2 r \sin(\theta)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2} e^{(-\lambda(r,t))}}{2 r}\right) d\theta + \left(-\frac{i \sqrt{2} e^{(-\lambda(r,t))}}{2 r \sin(\theta)}\right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2} e^{g}}{2 \frac{\partial}{\partial t} \lambda(r,t)} \partial_t - \frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \partial_r$$

$$\Delta = -\frac{\sqrt{2} e^{g}}{2 \frac{\partial}{\partial t} \lambda(r,t)} \partial_t + \frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \partial_r$$

$$\delta = \frac{\sqrt{2} e^{(-\lambda(r,t))}}{2 r} \partial_\theta + \frac{i \sqrt{2} e^{(-\lambda(r,t))}}{2 r \sin(\theta)} \partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2} e^{(-\lambda(r,t))}}{2 r} \partial_\theta - \frac{i \sqrt{2} e^{(-\lambda(r,t))}}{2 r \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{1}{2}\,\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right) + \frac{1}{2}\,\sqrt{2}e^{g} + \frac{\sqrt{2}e^{(-\lambda(r,t))}}{2\,r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2}\,\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right) - \frac{1}{2}\,\sqrt{2}e^{g} + \frac{\sqrt{2}e^{(-\lambda(r,t))}}{2\,r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)e^{(-\lambda(r,t))}}{4\,r\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)e^{(-\lambda(r,t))}}{4\,r\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\,\sqrt{2}e^{g} - \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda\left(r,t\right)}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

$$\gamma = \frac{1}{4}\,\sqrt{2}e^{g} - \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda\left(r,t\right)}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{2}\, e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{(\partial r)^2}\lambda\left(r,t\right) + \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{2\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,r} - \frac{e^{(2\,g)} \frac{\partial}{\partial t} g}{2\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} + \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{2\,r\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{4}\, e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)^2 - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,r} - \frac{e^{(2\,g)} \frac{\partial}{\partial t} g}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} + \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^3}{(\partial r)^2 \partial t}\lambda\left(r,t\right)}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{1}{2}\, e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{(\partial r)^2}\lambda\left(r,t\right) + \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{2\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,r} - \frac{e^{(2\,g)} \frac{\partial}{\partial t} g}{2\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} + \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{2\,r\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

$$\begin{aligned}\Lambda = &-\frac{1}{12}\, e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)^2 - \frac{1}{6}\, e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{(\partial r)^2}\lambda\left(r,t\right) - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{12\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} \\ &- \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial}{\partial r}\lambda\left(r,t\right)}{3\,r} + \frac{e^{(2\,g)} \frac{\partial}{\partial t} g}{4\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^3}{(\partial r)^2 \partial t}\lambda\left(r,t\right)}{12\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{\left(-2\,\lambda(r,t)\right)} \frac{\partial^2}{\partial r \partial t}\lambda\left(r,t\right)}{6\,r\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} + \frac{1}{2}\, e^{(2\,g)}\end{aligned}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = \frac{1}{6}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)^2 - \frac{1}{6}\, e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)\frac{\partial^2}{\partial r\partial t}\lambda\left(r,t\right)}{3\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)}{6\,r} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\lambda\left(r,t\right)}{6\,\frac{\partial}{\partial t}\lambda\left(r,t\right)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda\left(r,t\right)}{6\,r\,\frac{\partial}{\partial t}\lambda\left(r,t\right)}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"